

• 2.1

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

a.

$$\bar{x} = \frac{\sum x_i}{N} = 1 \quad \#$$

$$\bar{y} = \frac{\sum y_i}{N} = 2 \quad \#$$

b.

$$b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \quad (2.7)$$

$$\Rightarrow b_2 = \frac{8}{10} = 0.8 \text{ (slope)}$$

For every one-unit increase in x , y increases by 0.8 units. #

$$b_1 = \bar{y} - b_2 \bar{x} \quad (2.8)$$

$$\Rightarrow b_1 = 2 - 0.8 \times 1 = 1.2 \text{ (intercept)}$$

When x is zero, y is 1.2. #

c.

$$\bullet \sum_{i=1}^5 x_i^2 = 9 + 4 + 1 + 1 + 0 = 15$$

$$\text{Show that } \sum (x_i - \bar{x})^2 = \sum x_i^2 - N \bar{x}^2$$

$$\Rightarrow 10 = 15 - 5 \times 1^2 = 10 \quad \#$$

$$\bullet \sum_{i=1}^5 x_i y_i = 12 + 4 + 3 - 1 + 0 = 18$$

$$\text{Show that } \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N \bar{x} \bar{y}$$

$$\Rightarrow 8 = 18 - 5 \times 1 \times 2 = 8 \quad \#$$

d.

$$\hat{y}_i = 1.2 + 0.8 x_i$$

$$\hat{y} = 1.2 + 0.8 \times 1 = 2 \quad \#$$

$$\times \sum \hat{e}_i = 0 \text{ \& } \sum x_i \hat{e}_i = 0$$

符合 LSE 第一階段條件

$$y_i - \hat{y}_i$$

$$\bullet s_y^2 = \frac{\sum (y_i - \bar{y})^2}{N-1} = \frac{10}{4} = 2.5 \quad \# \text{ 變異性}$$

$$\bullet s_x^2 = \frac{\sum (x_i - \bar{x})^2}{N-1} = \frac{10}{4} = 2.5 \quad \# \text{ 變異性}$$

$$\bullet s_{xy} = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{N-1} = \frac{8}{4} = 2 \quad \# \text{ 共變異性}$$

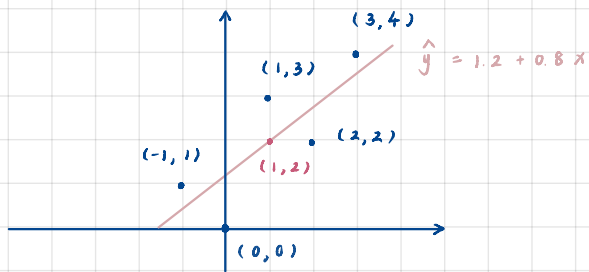
$$\bullet r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{2}{\sqrt{2.5} \sqrt{2.5}} = 0.8 \quad \# \text{ 相關係數}$$

$$\bullet CV_x = 100 \times \frac{s_x}{\bar{x}} = 100 \times \frac{\sqrt{2.5}}{1} = 158.11 \quad \# \text{ 變異係數}$$

$$\bullet \text{median} = 1 \quad \# \text{ 中位數}$$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

e.



g.

$$\bar{y} = 1.2 + 0.8x$$

plug $(1, 2)$ into the formula

$$\Rightarrow 2 = 1.2 + 0.8 \quad \#$$

h.

$$\hat{y} = \frac{\sum y_i}{N} = \frac{10}{5} = 2 = \bar{y} \quad \#$$

i.

$$\hat{\sigma}^2 = \frac{\sum (y_i - \hat{y})^2}{N-2} = \frac{3.6}{3} = 1.2 \quad \#$$

j.

$$\widehat{\text{var}}(b_2 | x) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12 \quad \#$$

$$\text{se}(b_2) = \sqrt{0.12} = 0.346 \quad \#$$

2.22

a.

- $\text{TOTALSCORE}_i = \beta_1 + \beta_2 \text{SMALL}_i + e_i$
- Interpretation:
 - $b_1(\text{intercept}) = 916.442 \rightarrow$ It represents the estimated average score on the achievement test for students in **regular-sized classes**.
 - $b_2(\text{slope}) = 12.175 \rightarrow$ It indicates the **estimated difference** in average test scores between students in small-sized classes and regular-sized classes. On average, students in small-sized classes score about **12.175 points higher**.
- Conclusion:
 - Based on the point estimates, class size appears to affect learning outcomes, with small-sized classes showing a positive impact on student performance.

b.

- $\text{READSCORE}_i = \beta_1 + \beta_2 \text{SMALL}_i + e_i$
- Interpretation:
 - $b_1(\text{intercept}) = 432.665$
 - $b_2(\text{slope}) = 6.924$
- $\text{MATHSCORE}_i = \alpha_1 + \alpha_2 \text{SMALL}_i + e_i$
- Interpretation:
 - $a_1(\text{intercept}) = 483.777$
 - $a_2(\text{slope}) = 5.251$
- Conclusion:
 - In regular-sized classes, students have higher average scores in math compared to reading.
 - However, the positive effect of small classes is larger for reading.

c.

- $\text{TOTALSCORE}_i = \beta_1 + \beta_2 \text{AIDE}_i + e_i$
- Interpretation:
 - $b_1(\text{intercept}) = 916.442 \rightarrow$ It represents the estimated average score on the achievement test for students in **regular-sized classes without an aide**.
 - $b_2(\text{slope}) = 4.306 \rightarrow$ It indicates the **estimated difference** in average test scores between students in regular-sized classes with an aide and those without an aide. On average, students in regular-sized classes with an aide score about **4.306 points higher**.

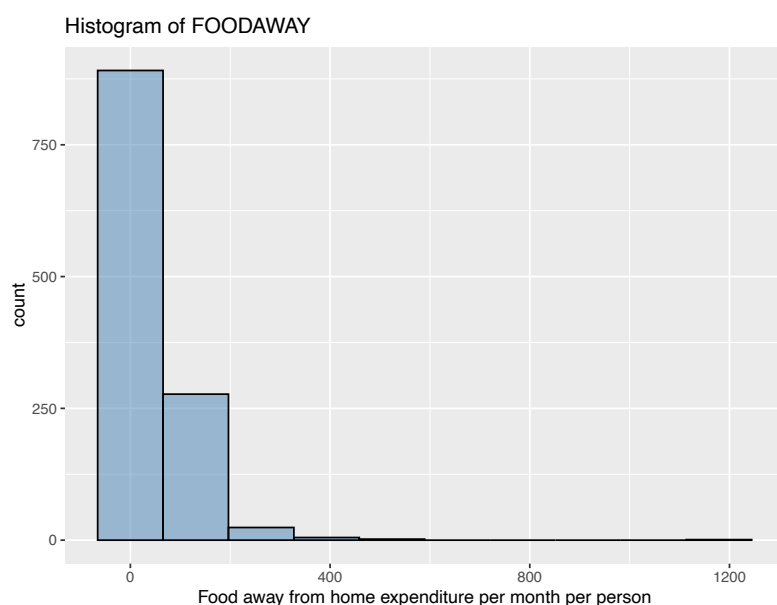
d.

- $\text{READSCORE}_i = \beta_1 + \beta_2 \text{AIDE}_i + e_i$
- Interpretation:
 - $b_1(\text{intercept}) = 432.665$
 - $b_2(\text{slope}) = 2.871$
- $\text{MATHSCORE}_i = \alpha_1 + \alpha_2 \text{AIDE}_i + e_i$
- Interpretation:
 - $a_1(\text{intercept}) = 483.777$
 - $a_2(\text{slope}) = 1.435$
- Conclusion:
 - In regular-sized classes, having an aide increases the average reading score by 2.871 points, compared to an increase of only 1.435 in math.

2.25

a.

- The histogram of FOODAWAY



- Summary:

- mean = 49.27
- median = 32.55
- 25th percentiles = 12.04
- 75th percentiles = 67.50

```
> # FOODAWAY's summary statistics
> summary(cex5_small["foodaway"])
      foodaway
Min.   : 0.00
1st Qu.: 12.04
Median : 32.55
Mean   : 49.27
3rd Qu.: 67.50
Max.   :1179.00
```

b.

- Households including a member with an advanced degree
- Summary:
 - mean = 73.15
 - median = 48.15
- Households including a member with a college degree
- Summary:
 - mean = 48.60
 - median = 36.11
- Households including a member with no advanced or college degree
- Summary:
 - mean = 39.01
 - median = 26.02

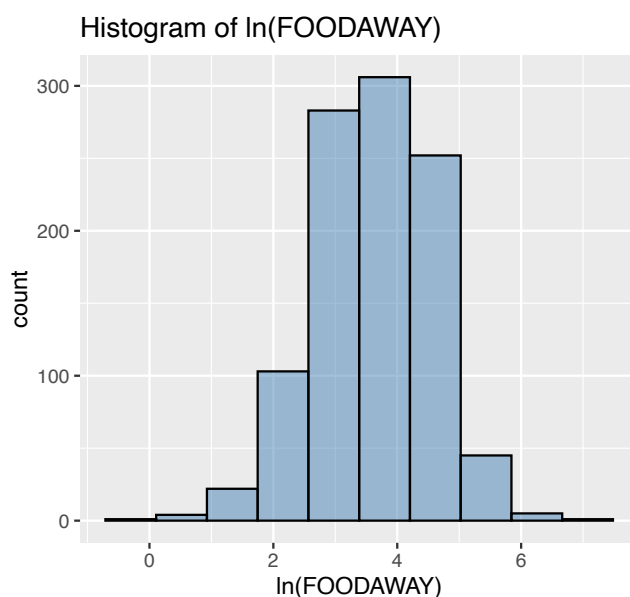
```
> summary(advanced_data["foodaway"])
foodaway
Min.   : 0.00
1st Qu.: 21.67
Median : 48.15
Mean   : 73.15
3rd Qu.: 90.00
Max.   :1179.00

> summary(college_data["foodaway"])
foodaway
Min.   : 0.00
1st Qu.: 14.44
Median : 36.11
Mean   : 48.60
3rd Qu.: 68.67
Max.   :416.11

> summary(no_data["foodaway"])
foodaway
Min.   : 0.00
1st Qu.: 9.63
Median : 26.02
Mean   : 39.01
3rd Qu.: 52.65
Max.   :437.78
```

c.

- The histogram of $\ln(\text{FOODAWAY})$



- Summary:
 - mean = -inf
 - median = 3.483
 - 25th percentiles = 2.488
 - 75th percentiles = 4.212

```
> summary(log(cex5_small$foodaway))
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 -Inf    2.488    3.483   -Inf    4.212    7.072
```

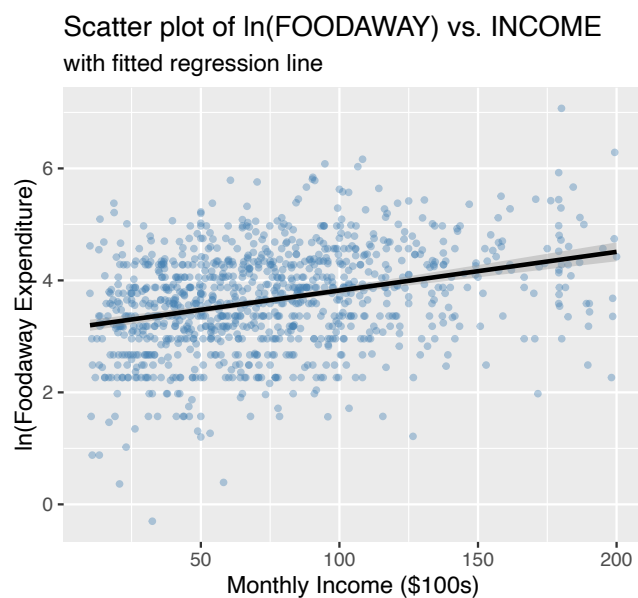
- Conclusion:
 - The number of observations for $\ln(\text{FOODAWAY})$ is **smaller** than that of FOODAWAY . Because the original data set contains households with **0 expenditure on foodaway**. Mathematically, the natural log of zero is undefined and approaches **negative infinity**.

d.

- $\ln(\text{FOODAWAY}) = \beta_1 + \beta_2 \text{INCOME}_i + e_i$
- Interpretation:
 - $b_2(\text{slope}) = 0.0069 \rightarrow$ It indicates that for every additional unit increase in monthly income, the monthly expenditure on food away from home is estimated to increase by **0.69%**.

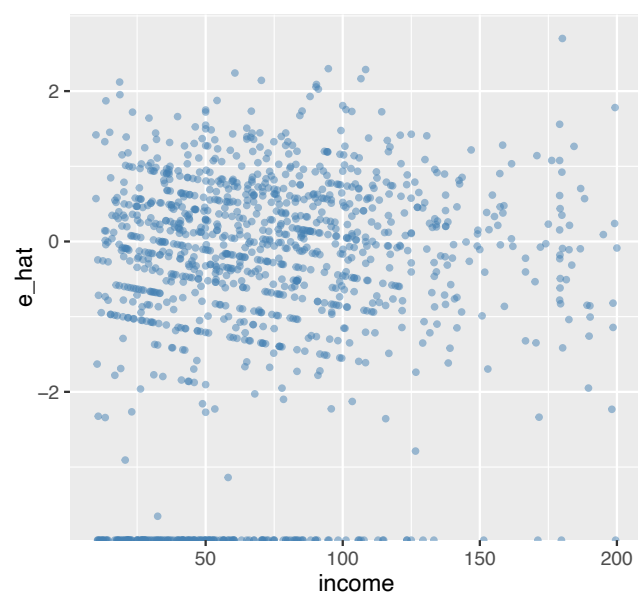
e.

- Scatter plot of $\ln(\text{FOODAWAY})$ vs. INCOME with a fitted line



f.

- Least squares residuals = $-\text{inf}$
- Scatter plot



- Conclusion:
 - The residuals are **randomly distributed** and show **no patterns**.
 - The log-linear model provides a reliable estimation of the relationship between income and food expenditure.